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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

IS3001-1 – DATA COMMUNICATION AND NETWORKING
(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following is a type of guided transmission media?
A) Radio waves B) Fiber Optics
C) Infrared D) Satellite
2. What is the main purpose of the data link layer?
A) To establish connections B) To provide error-free transfer of data frames
C) To manage sessions D) To route packets
3. What is the function of the network layer?
A) To manage data flow B) To encrypt data
C) To route packets between devices D) To establish sessions
4. Which protocol is associated with connection-oriented service?
A) TCP B) IP
C) UDP D) ARP
5. What is the role of the medium access control sublayer?
A) To manage the physical medium B) To establish connections
C) To control access to the shared medium D) To route data
6. Which of the following describes a "collision" in a network?
A) When two packets are sent simultaneously on the same channel B) When a device fails
C) When data is corrupted D) When a network is overloaded
7. Which of the following is a disadvantage of wireless transmission?
A) High speed B) Mobility
C) Flexibility D) Susceptibility to interference
8. What is the primary goal of network security?
A) To improve speed B) To prevent unauthorized access
C) To increase bandwidth D) To enhance performance
9. What should be the flag value to indicate the last fragment?
A) 0 B) 1
C) Protocol field value D) TTL value
10. In case of a collision in the channel between two stations, what follows?
A) Retransmit frame immediately B) Do nothing
C) Wait a Random time and retransmit frame D) Wait infinitely
11. The TTL field has value is 10. How many routers (max) can process this datagram?
A) 10 B) 1
C) 11 D) 5
12. The sizes of source and destination port address in TCP header are _____ respectively.
A) 16-bits and 32-bits B) 32-bits and 16-bits
C) 32-bits and 32-bits D) 16-bits and 16-bits
13. In classless addressing, there are no classes, but addresses are still granted in _____.
A) IPs B) Codes
C) Sizes D) Blocks

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14. Which of the following is false with respect to TCP?
 A) Connection-oriented B) Transport layer protocol
 C) Process to process D) Unreliable
15. In the IPv6 header, the traffic class field is similar to which field in the IPv4 header?
 A) Fragmentation field B) ToS field
 C) Fast-switching D) Open field
16. TCP and UDP are called _____.
 A) Transport protocols B) Session protocols
 C) Application protocols D) Network protocols
17. What is the primary purpose of the TCP segment header?
 A) Identifies routing paths B) Provides information for data transfer control
 C) Controls data formatting D) Encodes data for security
18. SMTP is used primarily for:
 A) Web browsing B) File transfer
 C) Sending emails D) Streaming video
19. What is the main purpose of DNS?
 A) Encrypting data B) Resolving domain names to IP addresses
 C) Managing email transfers D) Establishing network connections
20. Which protocol is commonly used to retrieve email from a server?
 A) POP3 B) SMTP
 C) FTP D) HTTP

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Explain the TCP/IP reference model, including all its layers. | 6 | L*2 | 1 | 1,2 |
| b) Let d = 00110001110, so k=11 and r= 4, H(15,11). Construct a 4X4 matrix and demonstrate how single-bit errors can be corrected in this Hamming code. | 10 | L3 | 1 | 1,2 |
| 2. a) List different types of guided transmission media. Also, explain the structures of twisted pair and coaxial cable. | 8 | L2 | 1 | 1,2 |
| b) Explain the working of a Simplex Stop-and-Wait protocol. Construct the scenarios of frame lost and acknowledgement lost with an example. | 8 | L3 | 1 | 1,2 |
| 3. a) Explain the hierarchical routing algorithm with an example. | 8 | L2 | 2 | 1,3 |
| b) Construct a frame using character and bit-oriented approaches and clearly illustrate stuffing and unstuffing in both cases. | 8 | L3 | 2 | 1,3 |

Unit – II

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|---|---|----|---|-----|
| 4. a) Explain the importance of fragmentation in the internetworking with the help of an example. | 8 | L2 | 3 | 1,3 |
| b) Discuss the role of tunnelling in transmitting packets using different network protocols. | 8 | L2 | 3 | 1,3 |
| 5. a) Explain the IPv4 datagram with its header format. | 8 | L2 | 4 | 1,2 |
| b) Explain various principal ICMP message types. | 8 | L2 | 4 | 1,2 |
| 6. a) Explain four scenarios of releasing a connection. | 8 | L2 | 4 | 1,2 |
| b) Explain with an example the process of recovery from host crashes. | 8 | L2 | 4 | 1,2 |

Unit – III

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|---|---|----|---|-----|
| 7. a) Discuss different fields of TCP header segment with its format. | 8 | L2 | 5 | 1,2 |
| b) Explain the different steps involved in a resolver for looking up a remote name. | 8 | L2 | 5 | 1,2 |
| 8. a) Explain the different MIME content types and example subtypes. | 8 | L2 | 5 | 1,2 |
| b) Explain the steps involved in accessing the following URL.
http://www.cs.washington.edu/index.html. | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
